

Harim Jin

Max Planck Institute for Astrophysics
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☆ [Stellar Astrophysics](#) ☆ [Supernova](#) ☆ [Data Science](#)

Work

Max Planck Institute for Astrophysics

Nov 2025–

Independent Postdoctoral Fellow

Mentor: [Selma E. de Mink](#)

Education

University of Bonn

Apr 2021–Dec 2025

Thesis: "Chemical fingerprints of stellar evolution in Galactic massive stars"

Graduation with cum laude

International Max Planck Research Schools (IMPRS) Fellow

Supervisor: [Norbert Langer](#)

Seoul National University

Mar 2019–Mar 2021

M.Sc. in Astronomy (99.1% – GPA 4.4 / 4.5)

Thesis: "Light Curve and Color Properties of Type Ib/Ic Supernovae"

Supervisor: [Sung-Chul Yoon](#)

Korea Advanced Institute of Science and Technology (KAIST)

Mar 2014–Feb 2019

B.Sc. in Physics (96.67% – GPA 4.0 / 4.3)

Presidential Fellow (top ~ 1 % of students)

Graduation with summa cum laude

Skills and Qualifications

Stellar Evolution Modelling

↳ Expert in detailed stellar evolution simulations using MESA

Supernova Modelling

↳ Proficient in simulating supernova explosions and subsequent kinematics and radiation evolution using STELLA

Statistical Data Analysis

↳ Proficient in applying Bayesian inference to large data sets of stellar models

Programming and Web Development

↳ Expert in Python, with proficiency in Fortran and Java; developed web services

Honors and Awards

- International Max Planck Research Schools (IMPRS) Fellowship** Apr 2021–Dec 2025
 ↳ Financial and academic support from Max Planck Institutes
- Scholarship for Excellent Academic Achievements** Sep 2020–Dec 2020
 ↳ Financial support for academically high-achieving students
- Fund from National Research Foundation (NRF) of Korea** Apr 2019–Mar 2021
 ↳ Financial support for master program
- KAIST Presidential Fellowship** Mar 2014–Feb 2019
 ↳ The highest honor for distinguished undergraduate students at KAIST
- Fulbright & GE Foundation Scholar-Leaders Program** Jun 2015–Feb 2019
 ↳ Scholarship award from Fulbright for potential contributions to society
 ↳ (awarded to only 5 undergraduate students in Korea)
- Dean's List of College of Natural Science** Sep 2015–Dec 2015
 ↳ Scholarship award for academic excellence
- National Scholarship for Science and Engineering** Mar 2014–Feb 2018
 ↳ Financial support of full tuition for 4 years of undergraduate studies at KAIST
- 2nd prize in the 9th APAO (Asian-Pacific Astronomy Olympiad)** Dec 2013
 ↳ The second-highest honor for achievements in the Olympiad

Teaching and Outreach

- Argelander-Institut für Astronomie (AlfA) Science Day** Jun 2024
 ↳ Organized by the DEI (diversity, equity, inclusion) committee of AlfA
 ↳ Introduced stellar astronomy to students
- Teaching Assistant, University of Bonn** Apr 2022–Feb 2024
 ↳ Conducted exercise classes/labs in "Stars and Stellar Evolution" (fall semester)
 ↳ Conducted exercise classes in "Nucleosynthesis" (spring semester)
- Teaching Assistant, Seoul National University** Sep 2019–Dec 2019
 ↳ Organized practical exercises in "Evolution of the Universe"
- Teaching Assistant, Seoul National University** Mar 2019–Jun 2019
 ↳ Assisted students with their writing skills in "Human and the Universe"
- Talented Education Program Tutor, KAIST** Apr 2018–Nov 2018
 ↳ Conducted physics lectures for high-achieving students
- Volunteer Tutoring Program, KAIST** Oct 2015–Dec 2015
 ↳ Taught calculus and physics to high school students

Publications (TBD)

First-author publications

1. **Harim Jin**; Selma E. de Mink; Sebastian Holzner; Jakub Klencki; Géza Csörnyei; Sung-Chul Yoon; Iair Arcavi; Wolfgang E. Kerzendorf (2026) [Type Ib Supernovae are bluer than Type Ic Supernovae](#). Submitted to A&A
2. **Harim Jin**; Norbert Langer; Andrea Ercolino, Selma de Mink (2026) [Comprehensive massive binary evolution model grid for our galaxy](#).
3. **Harim Jin**; Norbert Langer; Lennon, Daniel J.; Charles R. Proffitt (2024) [Boron depletion in Galactic early B-type stars reveals two different main sequence star populations](#). 2024A&A...690A.135J
4. **Harim Jin**; Yoon, Sung-Chul; Blinnikov, Sergei (2023) [Optical Color of Type Ib and Ic Supernovae and Implications for Their Progenitors](#). 2023ApJ...950...44J
5. **Harim Jin**; Yoon, Sung-Chul; Blinnikov, Sergei (2021) [The Effect of Circumstellar Matter on the Double-peaked Type Ic Supernovae and Implications for LSQ14efd, iPTF15dtg, and SN 2020bvc](#). 2021ApJ...910...68J

Second-author publications

6. Ercolino, A.; **Jin, H.**; Langer, N.; Dessart, L. [Mass-transferring binary stars as progenitors of interacting hydrogen-free supernovae](#)
7. Charles R. Proffitt; **Harim Jin**; Daflon, Simone; Lennon, Daniel J.; Norbert Langer; Cunha, Katia; Monroe, Talawanda (2024) [Boron Abundances in Early B Dwarfs of the Galactic Open Cluster NGC 3293](#). 2024ApJ...968....1P
8. Ercolino, Andrea; **Harim Jin**; Norbert Langer; Dessart, Luc (2024) [Interacting supernovae from wide massive binary systems](#). 2024A&A...685A..58E

Other publications

9. Xu, X. -T.; Schürmann, C.; Langer, N.; Wang, C.; Schootemeijer, A.; Shenar, T.; Ercolino, A.; Haberl, F.; Hastings, B.; **Jin, H.**; Kramer, M.; Lennon, D.; Marchant, P.; Sen, K.; Tauris, T. M.; de Mink, S. E. Populations of evolved massive binary stars in the Small Magellanic Cloud I: Predictions from detailed evolution models, [10.48550/arXiv.2503.23876](https://arxiv.org/abs/10.48550/arXiv.2503.23876)
10. **Harim Jin**; [Implications of boron surface abundances in massive stars](#) Bulletin de la Société Royale des Sciences de Liège, vol. 93, no. 3 - Proceedings of the 41st Liège International Astrophysical Colloquium "The eventful Life of massive multiples", held in Liège 15-19 July 2024, eds Y. Nazé & E. Bozzo, pp. 275-282 10.25518/0037-9565.12353

11. Martínez-Sebastián, Carlos; Simón-Díaz, Sergio; **Harim Jin**; Keszthelyi, Zsolt; Holgado, Gonzales; Norbert Langer; Puls, J (2025) [Helium enrichment in O-type stars as a tracer of past binary interaction](#). 2025A&A...693L..10M
12. Dessart, Luc; Gutierrez, Claudia P.; Ercolino, Andrea; **Harim Jin**; Norbert Langer (2024) [A sequence of Type Ib, IIb, II-L, and II-P supernovae from binary-star progenitors of varying initial separation](#). 2024A&A...685A.169D
13. Yoon, Sung-Chul; Jung, Moo-Keon; **Harim Jin**; Kim, Hyun-Jeong (2022) [Effects of wind mass-loss on the observational properties of Type Ib and Ic supernova progenitors](#). 2022IAUS..366...39Y
14. Rho, J.; Evans, A.; Geballe, T.; ..., **Harim Jin**; ... et al. (2021) [Near-infrared and Optical Observations of Type Ic SN 2020oi and Broad-lined Type Ic SN 2020bvc: Carbon Monoxide, Dust, and High-velocity Supernova Ejecta](#). 2021ApJ...908..232R

Conference Activities (TBD)

- [Talk and Poster](#), EAS 2026
- [Poster](#), IAU transients symposium
- [Talk](#), APRIM 2026
- [Talk](#), KAS Spring meeting 2026
- [Talk](#), Ringberg Postdoc Conference 2026
- [Talk](#), Ringberg supernova 2025
- POSYDON school 2025
- [TA](#), MESA summer school Leuven 2025
- CCA Stable mass transfer workshop 2025
- [Talk](#), BLOEM meeting 2025
- [Talk](#), Seminar at Cologne University 2025
- [Talk](#), International Annual Meeting of the German Astronomical Society, Sep 2024
- Lorentz Center Workshop: From discovery to a population: benchmarking stripped stars and companions, Jul 2024
- [Poster](#), Liège International Astrophysical Colloquium: The eventful life of massive star multiples, Jul 2024
- [Organizer](#), Bonn neutron star workshop, May 2024
- [Poster](#), Science Meets Business and Poster Session, Apr 2024
- Stable mass transfer workshop, Mar 2024
- [Talk](#), VFTS meeting in Munich, Mar 2023
- 102th Korean Astronomical Society Fall Meeting, Oct 2020
- [Talk](#), 101st Korean Astronomical Society Fall Meeting, Nov 2019
- [Poster](#), Fifty-One Erg Supernova Conference, May 2019
- [Poster](#), 10th DTA symposium “Stellar deaths and their diversity”, Jan 2019

- 99th Korean Astronomical Society Fall Meeting, Oct 2018
- 93rd KPS (Korean Physical Society) Spring Meeting, Apr 2017
- Talk, KAIST Long-term Undergraduate Research Program Workshop, Feb 2017
- 99th Korean Astronomical Society Fall Meeting, Oct 2018
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- Talk, KAIST Long-term Undergraduate Research Program Workshop, Feb 2017